

Automated Ambient Light Control and Actuation System for Smart Environments

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Abstract: This paper represents the architectural design and software of an autonomous smart indoors management system aimed at reducing energy waste while adding quality of life improvements. The edge computing internet of things solution utilizes an ESP32 microcontroller and a multi sensor fusion strategy, combining passive infrared (PIR), millimeter-Wave (mmWave) radar and an ambient illuminance monitoring for occupancy detection and ambient light regulator. Unlike binary systems, the proposed firmware uses a proportional position control algorithm via time-based calculation to dynamically adjust window shutters, alongside a dedicated privacy-focused "night state." Actuation is safely isolated utilizing relays and non-invasive 38kHz Infrared (IR) transmission for localized HVAC termination. To validate the C++ state machine prior to physical deployment, the architecture was modeled on the Wokwi Electronic Design Automation platform, with Bluetooth Low Energy (BLE) parameter configuration abstracted via UART serial injection. Simulated results confirm the system's capability in reducing the passive power draw while maintaining network-independent edge-computing reliability.

I-Introduction:

For large-scale educational institutions, energy consumption represents one of the highest continuous operational expenditures, with artificial lighting and heating, ventilation, and air-conditioning (HVAC) systems accounting for the vast majority of their electrical loads. A significant portion of this expenditure is driven by idle energy waste when high-current systems remain active in unoccupied rooms due to human negligence or inflexible scheduling systems. While the integration of the internet of things (IoT) has provided a baseline for mitigating energy waste across various building profiles, the distinct occupancy behaviors of educational facilities require a specialized, localized control architecture.

In residential environments, IoT architectures heavily prioritize visual comfort and human-centric interactions. Frameworks such as the GreenSoul project^[1] demonstrate that domestic and public-use spaces benefit from context-aware systems that adapt to occupant behavior rather than relying on aggressive, rule-based shutdowns. These systems utilize ambient feedback and localized edge devices to improve quality of life, adjusting lighting incrementally to match natural circadian

rhythms. However, applying this "soft-timeout" logic to high-load facilities results in prolonged, unnecessary HVAC usage when occupants temporarily leave a space.

Conversely, commercial office environments require more aggressive power management to offset their square footage. To address this, intelligent architectures like the plug-mate system^[2] have successfully mitigated passive power draw by deploying IoT networks that learn and react to individual employee occupancy patterns. By shifting away from static calendars to real-time, data-driven occupancy detection, commercial systems can automatically sever power to localized loads when a desk is empty. Yet, translating this highly individualized, desk-specific control logic to an educational facility is impractical. Classrooms represent shared, chaotic spaces characterized by unpredictable, high-density occupancy turnover, necessitating a broader sensor-fusion approach rather than individualized plug management.

Recent studies focusing specifically on educational facilities, such as the smart classroom architecture proposed by Maduwantha et al.^[3], successfully utilize IoT microcontrollers paired with Passive Infrared (PIR) and Light Dependent Resistors (LDR) to mitigate localized energy waste. However, these baseline systems present considerable hardware limitations. First, they are vulnerable to false-negative occupancy readings; PIR

sensors detect only macro-motion, meaning a system may erroneously plunge a room into darkness while students are sitting stationary during an examination. Second, they frequently rely on binary actuation (turning lights purely on or off) rather than utilizing changing available daylight. Finally, their reliance on university Wi-Fi and centralized cloud dashboards introduces latency and network-failure vulnerabilities.

To bridge this gap, this paper proposes a localized, edge-computing IoT architecture tailored for the classroom occupancy profile. Utilizing an ESP32 microcontroller, the system introduces a multi-tier sensor fusion approach combining PIR for macro-motion with Millimeter-Wave (mmWave) radar to continuously detect micro-movements, ensuring low false-negative shutdowns during quiet study periods. To maximize daylight harvesting, the firmware executes a proportional position control algorithm to dynamically actuate heavy window shutters, calculating precise physical positioning to maintain a targeted lux value, alongside a dedicated "night state" to eliminate exposure and ensure occupant privacy. By executing all logic on the edge and managing parameters via secure Bluetooth Low Energy (BLE), the architecture tries to give a continuous, robust HVAC and lighting automation completely independent of internet connectivity.

II-System Architecture & Methodology

To achieve autonomous, reliable environmental control without the vulnerabilities associated with cloud dependent systems, the proposed architecture relies entirely on localized edge execution. The system is structurally divided into a central processing tier, a sensor fusion input stage, and an isolated electromechanical actuation stage.

A. Central Processing and Edge Logic:

The core logic of the proposed system is governed by an ESP32 dual-core microcontroller. This localized approach reduces network latency and ensures that automated HVAC and lighting controls remain fully operational during university network outages.

To maximize computational efficiency, the firmware leverages the ESP32's dual-core architecture. One core is dedicated to maintaining a high-frequency polling loop for real-time sensor evaluation and proportional motor calculations. The secondary core manages a secure Bluetooth Low Energy (BLE) GATT server, permitting asynchronous parameter tuning without interrupting the primary control loop. To ensure hardware safety, the power distribution is tiered; a dedicated step-down buck converter provides a stable 3.3V and 5V logic supply to the microcontroller and sensors, safely isolating the sensitive logic channels from the high-current demands of the electromechanical actuators.

B. Sensor Fusion and Occupancy

Validation: Accurate occupancy detection is the critical foundation of any automated

energy management system. As previously mentioned, educational environments frequently suffer from "stationary occupant" false-negatives when relying solely on traditional sensing methods. To resolve this, the architecture employs a multi-tiered sensor fusion strategy.

Initial room entry (macro-motion) is detected using an HC-SR501 Passive Infrared (PIR) sensor, which provides an instantaneous hardware interrupt to transition the room from an unoccupied baseline. However, to maintain the active state during low-movement periods, such as an examination or a quiet lecture, an HLK-LD2410 24GHz mmWave radar is integrated into the logic array. The mmWave sensor emits high-frequency electromagnetic waves to detect micro-movements, including human respiration and typing, which fall below the detection threshold of standard PIR sensors.

In the firmware, these two sensors are unified via a deterministic logical OR gate (PIR || mmWave). If either sensor registers a HIGH state, the system resets a non-blocking timeout counter and locks the room into an "occupied" state. The room is only authorized to initiate its energy-saving shutdown sequence if a continuous duration of zero-motion is registered across both sensors for a pre-defined period (configured to 10 minutes for physical deployment).

Simultaneously, ambient light levels are continuously quantified using an integrated Light Dependent Resistor (LDR) array. To ensure the control loop responds to standardized physical units rather than arbitrary analog-to-digital conversion noise, the raw voltage divider output is mathematically mapped to a localized

illuminance value (Lux). This standardized telemetry provides the required parameters for the daylight utilization algorithms.

C. Proportional Actuation Control:

Traditional automated shading systems frequently rely on binary logic, fully deploying or retracting window treatments based on arbitrary thresholds. This approach fails to optimize daylight harvesting and often results in jarring lighting transitions for the occupants. To resolve this, the proposed firmware introduces a proportional kinematic control algorithm.

The system continuously evaluates the absolute difference between the mapped ambient LDR reading and the target illuminance. If this difference exceeds a predefined interval implemented to prevent constant motor oscillation during minor cloud cover, the algorithm calculates the exact physical correction required. The lux differential is converted into a physical travel distance (centimeters) using a measured hardware constant. This distance is subsequently divided by the electromechanical motor's travel speed (cm/s) to dynamically calculate a precise motor runtime in milliseconds. The microcontroller then engages the directional relays for this exact duration, allowing the heavy exterior window shutters to descend or ascend to a precise fractional coordinate, thereby harvesting maximum natural daylight while eliminating solar glare.

Furthermore, the logic incorporates a dedicated "night state" threshold. If ambient light drops below a certain limit, the system bypasses proportional calculations and fully deploys the shutters. This acts as a physical visual barrier, eliminating the 'fishbowl

effect' of an illuminated room at night to ensure occupant privacy.

To ensure the longevity of the ESP32 and comply with electrical safety standards, all high-voltage actuation is strictly isolated. The artificial lighting and bidirectional tubular motors are driven via optically-isolated 5V relays. Crucially, rather than intercepting the high-current mains wiring of the classroom's existing HVAC split-units, the system utilizes a 38kHz Infrared (IR) transmitter. Upon an occupancy timeout, the ESP32 transmits the corresponding hexadecimal shutdown code, acting as a localized, non-invasive digital remote control to sever the HVAC load.

D. BLE Abstraction and Dynamic Param:

A significant limitation of strictly edge-computed IoT devices is their rigidity; hard-coded parameters cannot account for the shifting seasonal conditions or varying room layouts of an educational facility. However, relying on a Wi-Fi dashboard for parameter adjustment reintroduces the network vulnerabilities this architecture seeks to avoid.

To balance edge reliability with facility-management scalability, the system integrates a Bluetooth Low Energy (BLE) abstraction layer. Utilizing the secondary core of the ESP32, the microcontroller acts as a secure GATT (Generic Attribute Profile) server. Facility managers can connect locally via a mobile terminal to asynchronously update non-volatile system variables in real-time. Configurable parameters include:

Target Illuminance (Lux): The baseline lighting requirement for the specific classroom.

Limit Steps (Interval): The acceptable lux deviation permitted before triggering motor actuation.

Critical Light Limit: The threshold denoting absolute darkness to trigger the privacy night state.

Occupancy Timeout: The exact duration of zero-motion required to initiate the HVAC and lighting shutdown sequence.

By utilizing BLE for configuration, the architecture remains adaptable for diverse educational environments while maintaining absolute isolation from external cloud networks.

III. Simulation Methodology and Hardware Abstraction

To validate the firmware's state-machine logic and closed-loop control algorithms prior to physical deployment, the system was prototyped using the Wokwi Electronic Design Automation (EDA) platform. This online simulation approach allowed for rigorous testing of the edge-computing parameters while bypassing the immediate electrical hazards of prototyping high-voltage relays and HVAC lines. Software-in-the-loop validation confirmed firmware functionality, while hardware performance remains subject to future experimental verification.

A. Component Abstraction and

Telemetry: Because EDA platforms do not inherently support every specialized physical sensor, specific hardware components were abstracted using deterministic equivalents to ensure the C++ control loop functioned identically to a physical build.

1. Occupancy Sensor Abstraction: While the standard HC-SR501 PIR sensor is natively supported in the simulation environment, the HLK-LD2410 mmWave radar requires abstraction. Because the mmWave radar outputs a standard 3.3V HIGH signal upon detecting micro-movements, it was simulated using a discrete Single-Pole Double-Throw (SPDT) digital slide switch. This allowed for the manual injection of sustained "micro-motion" HIGH states into the ESP32's GPIO, verifying the integrity of the deterministic OR gate array without requiring physical radar arrays.

2. Ambient Light and Logarithmic

Compensation: To simulate ambient daylight accurately, a virtual Light Dependent Resistor (LDR) was utilized. Because physical LDRs exhibit a logarithmic resistance curve relative to light intensity, standard linear software mapping introduces severe deviations at the lux extremes. To resolve this, the firmware incorporates a physics-based exponential compensation algorithm. The system continuously calculates the localized resistance from the raw 12-bit Analog-to-Digital (ADC) voltage and translates it using the GL5528 logarithmic curve. This ensures the proportional kinematic math receives highly accurate environmental telemetry rather than raw voltage approximations.

B. Accelerated State-Machine Validation:

Testing long-duration environmental timeouts in real-time is highly inefficient during the rapid prototyping phase. Therefore, the system's operational timescales were mathematically accelerated for the simulation.

The physical “occupancy timeout” designed to lock down the room and transmit the 38kHz HVAC shutdown signal after 10 minutes (600,000 ms) of zero-motion was scaled down to 10 seconds (10,000 ms). This acceleration successfully validated the shutdown sequence, confirming that the PIR and mmWave interrupts correctly reset the non-blocking timers without freezing the primary processor core.

Additionally, the Bluetooth Low Energy (BLE) parameter injection was simulated via Universal Asynchronous Receiver-Transmitter (UART) serial parsing. By transmitting string commands through the serial monitor, the simulation proved that facility managers could successfully overwrite non-volatile configuration variables (such as target illuminance and limit step deadbands) on the fly, dynamically altering the state machine's behavior without interrupting the localized edge-execution loop.

IV. Results and Discussion

To evaluate the efficacy of the proposed edge-computing architecture, the ESP32 firmware was subjected to a series of standardized operational scenarios within the simulation environment. The system's real-time Serial monitor telemetry was captured to verify the state-machine transitions, the proportional control math, and the safety isolations.

A. Scenario 1: Daylight Utilization and Proportional Management: The primary objective of the system is to dynamically maintain visual comfort without wasting energy. In this test scenario, the room was

triggered as occupied while the ambient exterior light was aggressively bright (800 Lux), far exceeding the system's 500 Lux target and 50 Lux deadband.

Recorded Telemetry:

System Booted.

Awaiting Sensor Input... --- ROOM

OCCUPIED --- [SYSTEM STATUS] Lux:

800 | Target: 500 | Occupied: YES |

Shutter Pos: 0.00 cm ACTUATOR: Relay

OPEN -> Artificial Lights OFF

ACTUATOR: Rolling DOWN 30.00cm.

Motor Runtime: 6000 ms. ACTUATOR:

Shutter Reached Target -> Power Cut.

[SYSTEM STATUS] Lux: 500 | Target: 500

| Occupied: YES | Shutter Pos: 30.00 cm

Discussion: The telemetry confirms the system correctly processed the non-linear analog inputs using the logarithmic GL5528 compensation algorithm to identify an accurate 800 Lux environment. From this true value, the logic calculated a precise 30.00 cm physical shutter correction (utilizing the 10 Lux/cm hardware constant). The motor runtime was mathematically derived as 6,000 milliseconds to achieve this exact deployment. Crucially, as the simulated room lux stabilized at the 500 Lux target, the system correctly severed power to the electromechanical relays, preventing

mechanical oscillation and maintaining the deadband without artificial lighting.

B. Scenario 2: The Privacy "Night State":

To validate the system's ability to override standard daylight harvesting in favor of occupant privacy, the ambient illuminance was artificially dropped below the threshold light limit while the room remained occupied.

Recorded Telemetry:

[SYSTEM STATUS] Lux: 45 | Target: 500 |

Occupied: YES | Shutter Pos: 30.00 cm

ACTUATOR: Relay CLOSED -> Artificial

Lights ON ACTUATOR: Rolling DOWN

170.00cm. Motor Runtime: 34000 ms.

ACTUATOR: Shutter Reached Target ->

Power Cut. [SYSTEM STATUS] Lux: 45 |

Target: 500 | Occupied: YES | Shutter

Pos: 200.00 cm

Discussion: The ESP32 successfully bypassed the proportional mathematics upon detecting a sub-100 Lux environment. It immediately closed the artificial lighting relays to illuminate the classroom and calculated the exact remaining distance (170.00 cm) required to fully deploy the shutters to their physical maximum. This validation ensures the classroom eliminates the illuminated "fishbowl effect" for occupants at night, prioritizing localized privacy over dynamic harvesting.

C. Scenario 3: Occupancy Timeout and Non-Invasive Shutdown:

The most important energy-saving function of the architecture is the autonomous shutdown sequence. In this validation, both the simulated PIR and mmWave inputs were driven LOW to simulate an empty classroom, triggering the accelerated 10-second timeout counter.

Recorded Telemetry:

[SYSTEM STATUS] Lux: 500 | Target: 500

| Occupied: NO | Shutter Pos: 200.00 cm

--- OCCUPANCY TIMEOUT REACHED ---

ACTUATOR: Relay OPEN -> Artificial

Lights OFF ACTUATOR: Transmitting

38kHz IR -> HVAC OFF

Discussion: The simulation verifies the non-blocking nature of the state machine; the system continues to output its telemetry heartbeat while processing the occupancy timeout. Once the threshold was crossed, the firmware executed the shutdown sequence. The lights were disengaged, the room was physically secured via shutter deployment, and the 38kHz infrared transmission command was dispatched. This confirms the system can safely isolate high-current HVAC loads without requiring direct, high-voltage mechanical interception by the microcontroller.

V. Conclusion

This paper presented the design and simulation of a localized, edge-computing IoT architecture tailored specifically for the energy management of educational facilities. By transitioning away from rigid scheduling and cloud dependencies, the proposed ESP32-based system provides an autonomous, real-time response to dynamic classroom environments. The integration of multi-tier sensor fusion combining PIR macro-motion detection with mmWave micro-motion validation—successfully lowers the "stationary occupant" false-negative vulnerabilities prevalent in legacy smart-building systems. Furthermore, the implementation of proportional kinematic shading algorithms maximizes natural daylight harvesting while maintaining strict occupant privacy through a dedicated visual night state.

Validation through Software-in-the-Loop simulation confirmed the integrity of the non-blocking state machines and the security of the BLE-based parameter configuration layer. By utilizing isolated electromechanical relays and non-invasive 38kHz infrared transmission, the architecture maintains safe operation without requiring direct interception of high-voltage HVAC mains.

While this framework was explicitly engineered to address the high-density, unpredictable occupancy profiles of university classrooms, the underlying edge-computing logic is acceptably scalable. Future implementations could adapt this sensor-fusion architecture for corporate offices to eliminate their reliance on centralized cloud servers, or scale it down for residential deployments where enhanced data privacy and localized

execution are prioritized. Ultimately, this localized approach provides a robust blueprint for mitigating unnecessary standby consumption across modern infrastructure.

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